PROPRA

MASTER SPEC ADDENDUM v3.1

Full Automation, Missing Screens & Zero-Human-Touch Specification

This document adds everything missing from v3.0: the complete zero-touch automation model, planning permission workflow, legal & conveyancing, procurement, marketing brochure generation, mapping & satellite, rendering visuals, transaction room, and explicit definitions of what the system does automatically with no human involvement at any step.

The most important sentence in this document

At no point in any Propra workflow does a human at Propra do anything on behalf of the user. Every quote is machine-generated from a live API. Every switch is executed by Propra software calling a supplier API. Every document is generated by AI. Every notification is automated. The word "we called" or "we contacted" never appears anywhere in Propra. Propra is software. It does not make phone calls. It does not send emails on behalf of the user asking for quotes. It calls APIs and returns structured data.

A The Zero-Touch Automation Model

What happens automatically the moment a property address is entered — no user action required beyond that

Principle

When a user adds a property by address, Propra immediately and automatically begins fetching data from every relevant API in parallel. By the time the user has finished the onboarding flow (under 3 minutes), their dashboard is populated with real data. Nothing is manual. Nothing requires a setup wizard. Nothing requires Ian.

A.1 Automatic data fetch on property address entry

Trigger	What system does automatically	What appears on screen	Speed	APIs called
Address entered	Geocodes address, fetches satellite imagery, loads boundary polygon, overlays planning layer, flood risk zone, EPC rating	Interactive satellite map with boundary, planning overlay, flood zone, EPC badge	**<5s**	*Google Maps Platform, Mapbox, OS NGD, EA Flood API, MHCLG EPC*
Address confirmed	Fetches current planning applications, decisions, appeal history, permitted development rights assessment	Planning history timeline, PD rights summary, any live applications	**<10s**	*Planning Portal API, LPA feeds, PINS API*

Property type confirmed	Pulls rental comparables, cap rate benchmarks, vacancy rates for that asset class + location	Market context panel: avg rent, cap rate, vacancy for comparable assets	**<8s**	*CoStar API, EG PropertyLink*
Insurance schedule uploaded	OCR extracts policy details, simultaneously calls all carrier APIs for replacement quotes	Three binding/indicative quotes with saving vs current. No call. No email.	**<60s**	*AWS Textract + AIG + Chubb + Lloyd's + Travelers + Aviva APIs*
Utility bill uploaded	Extracts tariff and usage, calls all energy supplier APIs for best available rate	Live tariff comparison: current vs best available, projected annual saving	**<30s**	*Textract + Octopus + EDF + British Gas + E.ON APIs*
Lease uploaded	Extracts all lease terms, calculates rent vs ERV, sets countdown timers, scores tenant leverage	Lease summary card, Rent Clock entry, leverage score with explanation	**<45s**	*Claude API + CoStar tenant data + D&B Hoovers*
Loan details entered or extracted	Fetches live SOFR rate, calculates DSCR, compares rate vs market, triggers lender API for indicative terms	Refinance opportunity card if gap exists, DSCR gauge, lender options	**<15s**	*FRED API + Trepp + lender panel APIs*
Property confirmed	Runs full AI opportunity scan across all categories: rent, insurance, energy, ancillary, planning, refi	AI Opportunity Centre populated with ranked list, total unactioned value shown	**<90s**	*Claude API + all category APIs*

A.2 What zero-touch means for each category

Category	Old world (what users do today)	Propra automated flow	Human involvement in Propra
Insurance	*Call broker. Wait 3-5 days. Broker calls carriers. Quotes arrive by email. Owner tries to compare PDFs.*	Upload schedule. Propra extracts details. Calls AIG/Chubb/Lloyd's/Travelers/Aviva APIs simultaneously. Returns structured quotes ranked by saving within 60 seconds. Click to bind. Document issued automatically.	**ZERO. No Propra employee touches this.**
Energy	*Call energy consultant or broker. Provide meter details. Wait for quote spreadsheet. Compare manually. Switch takes weeks.*	Upload bill or connect smart meter. Propra extracts tariff. Calls Octopus/EDF/British Gas APIs. Returns best rate. Click confirm. Propra executes the switch via supplier API. Confirmation in inbox.	**ZERO.**

Refinance	*Call mortgage broker. Provide financials. Broker shops the market. Multiple credit checks. Weeks of back and forth.*	Enter or upload loan details. Propra fetches live SOFR. Calculates DSCR. Sends automated DIP request to pre-vetted lender panel. Returns indicative terms. Owner selects. Propra facilitates completion digitally.	**ZERO on initial assessment. Human solicitor involved at legal completion only - standard conveyancing requirement.**
Planning	*Hire planning consultant. Wait for appraisal. Commission renderings separately. Submit application manually.*	Address loaded. Propra assesses PD rights automatically. AI generates planning appraisal. Renders proposed scheme via rendering API. Owner reviews. Propra submits via Planning Portal API. Status tracked automatically.	**ZERO. AI does the appraisal. API submits the application.**
Legal / Conveyancing	*Instruct solicitor by phone. Solicitor orders searches manually. Weeks of email chains.*	Owner triggers from Transaction Room. Propra instructs solicitor panel via API, pre-populating full property data pack. Searches ordered automatically. Results returned digitally. Drafted documents via AI.	**Solicitor does legal work (required by law). Propra removes all the admin around it.**
Procurement	*Get three quotes by phoning contractors. Wait days. Receive different-format quotes. Try to compare.*	Work order raised. Propra generates scope document. Sends to vetted contractor panel via API. Quotes returned in standard format. Owner selects on screen. Instruction issued. Job tracked.	**ZERO procurement admin. Contractors do the physical work.**
Marketing brochure	*Instruct agent or designer. Provide photos, floor plans, financials. Wait days. Pay £500-2000.*	Click Generate Brochure. Propra assembles satellite image, boundary, floor plan (if uploaded), financial summary, comparable evidence, AI investment narrative. Branded PDF ready in under 60 seconds.	**ZERO. AI writes it. APIs supply the data. Template is automatic.**
5G / EV / Solar	*Contact telecoms company. Wait for site survey appointment. Separate process for each income stream.*	Propra assesses suitability automatically using CTIL/Ofgem/Google Solar APIs. Shows	**ZERO on assessment and application. Physical installation by operator.**

		income potential. Owner approves. Propra submits application to operator/installer. Revenue share agreement auto-generated.	
Acquisitions	*Monitor Rightmove/CoStar manually. Call agent for details. Wait for IM. Commission own analysis.*	Propra monitors all portals 24/7. AI scores every new listing against portfolio criteria. Deal card appears when match found with full underwriting and income opportunity analysis. Owner acts or passes.	**ZERO monitoring. AI does the initial underwriting.**

B How Savings Are Calculated

Exact methodology — where every number comes from, how it is computed, what appears on screen

Principle

Every saving shown in Propra is calculated from real data — not estimated, not illustrative, not a range from a brochure. When Propra says you are overpaying £34,200 on insurance, it is because the live carrier APIs returned quotes totalling that difference right now. The user can see the source data behind every number.

Saving type	Inputs	Benchmark source	Calculation	Displayed as
Insurance saving	Current premium extracted from uploaded schedule (AWS Textract)	Live quotes from AIG, Chubb, Lloyd's, Travelers, Aviva APIs	Current premium minus cheapest live binding/indicative quote	**"Cheapest quote saves £X,XXX/yr. Click to bind." Carrier name shown.**
Energy saving	Current unit rate + standing charge from uploaded bill or smart meter	Live tariff from Octopus/EDF/British Gas/E.ON APIs for same meter type and consumption profile	Current annual spend (rate x kWh) minus best available annual rate x same kWh	**"Switch to [Supplier] saves £X,XXX/yr. Confirm to switch." Shows payback if any exit fee.**
Rent uplift	Passing rent from uploaded lease	CoStar comparable lettings for same asset type, location, specification within 12 months	ERV from comparables minus current passing rent, multiplied by floor area	**"£X,XXX/yr below market. Tenant leverage score: 7/10. Draft renewal letter."**
Refinance saving	Current loan rate + balance from uploaded mortgage schedule	Live SOFR from FRED API + lender panel indicative rates	Annual interest at current rate minus annual interest at best available rate on same balance	**"Refinancing to X% saves £X,XXX/yr. Lender terms available."**

CAM recovery	Service charge billed vs lease service charge cap + recoverable items from lease	Lease document (extracted) + standard recoverable items checklist	Identify items billed but not in lease, items under-recovered against cap	**"£X,XXX under-recovered this year. Reconciliation statement ready."**
Ancillary income	Roof area (from OS / satellite boundary), car park spaces (satellite), location signals (CTIL coverage maps)	CTIL rent benchmarks, Ofgem solar generation estimates, JCDecaux location CPM	Income per stream based on benchmark rates for asset characteristics	**"£X,XXX/yr available across 3 income streams. Tap to activate each."**
5G mast	Address, roof/land area, proximity to existing mast network (CTIL API)	CTIL current market rent by area and height	Market rent for site characteristics minus any existing income	**"Site qualifies for 5G mast. Market rent: £X,XXX-X,XXX/yr. Submit application."**
EV charging	Car park spaces, grid capacity (DNO API by postcode), daily footfall estimate	Pod Point / Osprey revenue share benchmarks by location type	Revenue share percentage x projected daily charge events x average session value	**"X spaces viable for EV charging. Projected: £X,XXX/yr. Install at zero cost."**
Solar	Roof area (satellite), aspect/orientation (OS mapping), current electricity rate	Google Maps Solar API for estimated generation, FIT/SEISS benchmark rates	Annual generation kWh x self-consumption saving + any export income	**"68,000 sqft roof. AI: 4.2yr ROI. Install at £0 upfront available."**

C Planning Permission — Full Workflow

From automatic assessment to rendered visuals to submitted application

What this is NOT

A page that shows planning application status from the portal. That is a monitoring tool. This is a full planning workflow that assesses what is possible, generates the analysis, creates visual renderings of proposed schemes, and submits applications — all via software.

C.1 Automatic planning assessment on property add

- Propra fetches current planning history and decisions from Planning Portal API and local authority feeds
- AI analyses the planning history, current use class, and surrounding context
- Permitted development rights assessed against current GPDO (England) or equivalent — what the owner can do without full planning permission
- Outputs: PD rights summary, any prior approval requirements, full planning needed/not needed
- All of this happens automatically when an address is added — no user request needed

C.2 Planning appraisal report — AI generated

- Owner selects a proposed use or change they are considering (e.g. "Convert ground floor from E class to residential")

- Propra AI generates a full planning appraisal: policy context, local plan policies, recent decisions on comparable applications, officer report tone analysis, likely outcome assessment
- Report is branded PDF, ready to share with planning consultant or use as briefing document
- Comparable decisions: Propra pulls recent planning decisions for similar applications in the same LPA to calibrate success probability
- This does not replace a qualified planning consultant for complex applications — but it replaces the initial £3k-£5k feasibility report

C.3 Rendering visuals — automated

Rendering type	What Propra generates	APIs / tools
Satellite context view	Satellite image with property boundary, proposed scheme footprint overlaid, neighbouring properties shown for context. Annotations showing what is proposed.	*Google Maps Platform (Static Maps API + Satellite layer), Mapbox GL JS, OS National Geographic Database for boundary*
Street-level before/after	Google Street View current state alongside AI-generated after view showing proposed facade, signage, or use change	*Google Street View Static API (before) + Stable Diffusion / DALL-E API for AI rendering (after) conditioned on current image*
Floor plan visualisation	If floor plan uploaded: proposed layout with new use annotated. If not uploaded: AI-estimated layout based on footprint and use class	*Claude API for layout reasoning + PDF floor plan extraction + Mapbox for floor plan rendering*
Planning application drawings	Basic location plan (1:1250) and site plan (1:500) auto-generated to standard planning requirements using OS mapping data	*Ordnance Survey API (location and site plan generation to planning standard scales)*
Massing model	For extension or development proposals: simple 3D massing shown against neighbouring properties to demonstrate scale	*Three.js (browser-rendered 3D) + OS building heights data + proposed parameters*

C.4 Application submission

- Full planning applications submitted via Planning Portal API — owner reviews pre-populated form, confirms, Propra submits
- Prior approval applications: simpler workflow, Propra auto-fills from property data and proposed use
- Status tracking: live updates from Planning Portal API — validated, valid, awaiting decision, approved, refused
- Appeal workflow: if refused, Propra generates appeal grounds document using AI analysis of refusal reasons vs comparable allowed appeals at PINS

D Legal & Conveyancing

Digital instruction, automated searches, AI-drafted documents — no phone calls

D.1 Automated legal instruction workflow

- From Transaction Room: owner confirms sale or purchase intent — Propra auto-populates legal instruction pack with all known property data (title, tenancies, incumbrances, planning history, searches already obtained)

- Propra API instructs pre-vetted solicitor panel (e.g. Cluttons Legal, SearchFlow panel firms) — no phone call, no email thread to start the process
- Solicitor receives structured data pack via API — not a phone call or email with a PDF attachment
- Progress tracked in Transaction Room: instruction received, searches ordered, searches returned, contract issued, exchange, completion

D.2 Automated property searches

Search type	Source	How ordered	Returned as
Local Authority search	Local Planning Authority	Search Acumen API or TM Group API — automated order on instruction	Structured JSON + PDF. Results parsed into key issues panel.
Water & drainage	Water company (e.g. Thames Water, Anglian)	CON29DW via InfoTrack or TM Group API	Flood risk, public sewer proximity, water supply type — structured data
Environmental search	Groundsure / Landmark / Argyll Environmental	Groundsure Avail API	Contamination risk, flood, radon, energy — parsed into risk flags
Land Registry search	HM Land Registry	HMLR API (official channels)	Title register, charges, registered proprietor — structured
Chancel repair search	Chancel repair liability register	SearchFlow or TM Group	Liability yes/no with indemnity insurance option if yes
Mining search	The Coal Authority, various	The Coal Authority API + regional search providers	Risk classification, historical workings data

D.3 AI lease and legal document drafting

- Heads of Terms: Claude API generates from deal parameters entered by owner (rent, term, breaks, incentives, permitted use) — formatted to RICS standard
- Lease renewal letter: generated from rent clock data — current rent, ERV, proposed new terms, break dates, strategy recommendation
- Section 25 notice (landlord): AI drafts from lease data when landlord does not wish to renew. Owner reviews and countersigns digitally.
- Section 26 request (tenant): if tenant requests renewal, Propra flags and prepares landlord response options
- Licence to assign / underlet: AI draft from standard precedent + specific lease terms extracted
- All AI-drafted documents reviewed by owner before any action — Propra does not send legal documents without owner confirmation
- Solicitor instruction: when document needs formal legal execution, Propra instructs panel solicitor via API with pre-populated draft

E Procurement & Contractor Tendering

Automated tender, structured quotes, digital instruction — no phone calls

E.1 How procurement works in Propra

- Work order raised by owner (or flagged by compliance tracker): description, property, urgency, estimated value
- Propra AI generates scope of works document from the description — structured, with standard clauses, H&S requirements, access instructions
- Scope sent to vetted contractor panel via API — panel organised by trade, geography, Propra rating
- Contractors submit quotes in Propra standard format (not PDF emails) — structured data: total, breakdown, start date, duration, warranty
- Owner sees quotes side-by-side on screen — not as separate PDF emails to interpret manually
- Owner selects: instruction issued automatically via API. Contractor receives formal instruction with scope and commercial terms.
- Job tracked: contractor updates status (mobilised, in progress, snagging, complete). Photos uploaded via contractor app.
- Completion: owner approves on screen. Payment instruction generated. Propra earns 3% of contract value.

E.2 Contractor panel — data requirements

- Every panel contractor: Companies House verification (API), PLI and employers liability insurance verified (API), RAMS templates on file, previous job ratings in Propra
- Propra does not add contractors who are not API-verified — no manual onboarding of contractors by an admin
- New contractor applications: self-serve form, automated Companies House check, insurance certificate OCR extraction, approval workflow

F Marketing Brochure & Investment Memorandum Generator

Branded professional property documents in under 60 seconds — no designer, no agent

What this replaces

A commercial agent charges £500-£2,000 and takes 3-5 days to produce a marketing brochure. An investment memorandum for a sale takes longer. Propra generates both in under 60 seconds using live data, satellite imagery, and AI-written narrative. Owner clicks Generate. Document is ready.

F.1 Marketing brochure — auto-assembled components

Section	Content	Source
Cover image	Aerial satellite image of property with boundary highlighted, professional crop and contrast	*Google Maps Platform Satellite Static API + boundary from OS NGD / Regrid*
Street view	Street-level photo of property frontage	*Google Street View Static API*
Property details	Address, tenure, use class, floor area, year built, EPC rating, planning history summary	*OS data, MHCLG EPC API, Planning Portal, lease register*
Location map	Annotated map showing property, transport links, nearby amenities, catchment area	*Mapbox GL JS with custom styling, Google Places API for amenity labels*
Financial summary	Current passing rent, net initial yield, total	*Live from Propra data — lease register, accounting*

	sq ft, estimated rental value, occupancy, NOI	*sync, CoStar ERV*
Tenancy schedule	All tenants, lease terms, rent, expiry dates, break dates — formatted as standard property schedule	*Lease register from Propra*
AI investment narrative	Two-page investment summary: location, asset quality, income security, value-add opportunity, market context	*Claude API with full property + market context*
Comparable evidence	Recent lettings and sales of comparable properties with achieved rents and yields	*CoStar API comparables for same asset class + geography*
Floor plans	If uploaded: formatted floor plans with areas annotated. If not: omitted.	*Uploaded PDF processed via Textract + formatted*

F.2 Output formats

- Branded PDF: Propra design template (or owner custom branding if provided) — download, share, or send directly from Propra
- Investment memorandum: Extended version with financial modelling, scenario analysis, comparable transactions, market commentary — same generation process, longer AI prompt
- Data room pack: All relevant documents (title, leases, EPC, planning, searches) bundled with brochure for buyer/investor due diligence — generated automatically from documents already in Propra
- API: Adobe PDF Services API for final PDF generation and formatting

G Mapping & Satellite — Full Specification

What loads automatically, what layers are shown, what the user can interact with

G.1 Automatic mapping on property add

- Every property gets a full interactive map view the moment the address is confirmed — no user request needed
- Default view: satellite with boundary polygon, annotated with address and key stats
- User can toggle layers: street map, satellite, planning overlay, flood risk, EPC heat map, comparable properties, transport

G.2 Map layers — full specification

Layer	What it shows	Data source	Trigger
Satellite base	High-resolution aerial image	Google Maps Platform Satellite / Mapbox Satellite	Always on by default
Property boundary	Red boundary polygon of the registered title	HM Land Registry / OS National Geographic Database / Regrid (US)	Auto on property add
Planning layer	Planning application pins with status colour	Planning Portal API + Historic England API + LPA GIS	Auto load, toggleable

	(approved/refused/pending), conservation areas, listed buildings	feeds	
Flood risk	Flood zone shading (1/2/3 for UK, A/B/C for US) overlaid on satellite	Environment Agency Flood API (UK), FEMA NFHL API (US)	Auto load, toggleable
EPC heat map	EPC ratings shown for surrounding properties — colour coded A-G	MHCLG EPC Register API	On-demand, toggleable
Comparables	Pins for recent comparable lettings/sales with rent/price labels	CoStar API comparables within 0.5 mile radius	On-demand, toggleable
Transport	Train stations, bus stops, motorway junctions, airports within relevant radius with travel times	Google Maps Distance Matrix API + TfL API (London)	On-demand, toggleable
Demographics	Population density, workforce size, income levels for catchment area	ONS API (UK), US Census Bureau API (US)	On-demand, toggleable
5G coverage	Mobile network coverage by operator, signal strength by location	CTIL coverage API, Ofcom Sitefinder API	On ancillary revenue screen
Grid capacity	Available electricity grid capacity by grid reference — for EV and solar viability	Distribution Network Operator APIs (UK), utility commission APIs (US)	On ancillary revenue screen
Street view	Ground-level photographic view, date-stamped, street position adjustable	Google Street View Static API / Mapbox Panoramax	On-demand panel

H Transaction Room

Full digital deal management — sale, purchase, or refinancing — from heads of terms to completion

Purpose

A commercial property transaction currently involves 15+ parties exchanging information via email over 3-6 months. Propra brings all parties into one digital room: owner, solicitor, agent, buyer, lender. Every document, every status update, every approval — in one place. Propra tracks it. Propra chases it. Nothing falls through the cracks.

H.1 Transaction types supported

- Sale of investment property: from instruction to agent/buyer through to completion
- Purchase of new asset: from identification through due diligence to completion
- Refinancing: from lender selection through legal charge to drawdown
- Lease renewal / regear: from heads of terms through to new lease execution

H.2 Transaction Room components

- Deal overview: parties, asset, deal type, key dates, status
- Document vault: all transaction documents versioned and party-accessible (NDA-gated for buyers)
- Data room: due diligence pack auto-assembled from Propra data (title, leases, EPC, planning, searches, insurance, financial summary)
- Milestone tracker: offer received, heads agreed, solicitors instructed, searches ordered, searches received, contract issued, exchange, completion — live status from API feeds where available
- NDA workflow: buyer receives NDA link, signs digitally (DocuSign or Docusign alternative API), access granted automatically
- Offer management: multiple offer tracking, accept/decline, counter-offer, best and final
- Communication log: all party communications logged — not replacing email but capturing outputs in one place

H.3 Integration requirements

- DocuSign API or HelloSign/Dropbox Sign API — digital signature for NDAs, contracts, legal documents
- SearchFlow / TM Group / InfoTrack — search ordering from within Transaction Room
- HMLR API — land registry updates, AP1/TR1 form generation
- Lender portal APIs (where available) — for refinance deal tracking
- Claude API — due diligence query answering from data room documents

---

I Project Intelligence

CapEx planning, contractor benchmarking, project P&L — all in one place

I.1 What Project Intelligence covers

- Capital expenditure planning: per-property CapEx schedule with cost estimates from contractor benchmarks
- Project P&L: budget vs actual, variance tracking, cost-to-complete
- Contractor benchmarking: cost per square foot for refurbishment, fit-out, mechanical, electrical by trade — benchmarked against Propra panel pricing and BCIS (Building Cost Information Service)
- Programme tracking: Gantt-style timeline with key milestones, contractor dependencies, critical path
- Defects tracker: snagging list with photo evidence, contractor response tracking
- EPC improvement pathway: cost vs rating improvement for each property — which works, what they cost, how much they improve the EPC, which tenants or requirements trigger the need

I.2 Integration requirements

- BCIS (Building Cost Information Service) API — industry standard construction cost benchmarks
- RIBA Plan of Work integration — project stage framework
- Companies House API — contractor verification within project context
- Glenigan API — construction project intelligence, market activity
- Google Maps Platform — site access routes, logistics planning

---

J Tenant Portal

Tenants interact with Propra directly — rent payments, maintenance, correspondence, document signing

J.1 Tenant portal features

- Rent payment: direct debit or card payment via Propra — confirms to Propra ledger automatically
- Maintenance requests: tenant raises via portal, Propra creates work order, assigns to contractor, tracks to completion — tenant receives automated updates
- Documents: lease, rent statements, maintenance records — tenant can view and download
- Communications: all landlord-tenant correspondence logged and sent via Propra — not personal email
- Digital document signing: lease renewals, licences, deeds — signed via DocuSign API from portal
- Rent account: statement of account, payment history, arrears if any

J.2 Integration requirements

- Stripe API or GoCardless API — rent payment processing (direct debit preferred for commercial rent)
- DocuSign API — document signing
- SendGrid / Resend — automated notification emails to tenants
- Open Banking / TrueLayer — payment confirmation against ledger

K Valuation & AVM

Automated property valuation on every asset — updated monthly

K.1 How automated valuation works in Propra

- On property add: AVM runs immediately using ATTOM (US) or CoStar AVM (UK/US) — returns estimated market value, confidence score, comparable sales used
- Monthly refresh: AVM runs automatically on all portfolio assets — updated value appears on dashboard
- Valuation history: chart showing how each property value has moved over time
- Yield calculation: auto-calculated from AVM value + NOI from accounting sync
- Valuation report: branded PDF export using AVM data + CoStar market commentary + Propra portfolio context — replaces £1k-£3k formal desktop valuation for portfolio monitoring purposes
- Caveat shown clearly: AVM is for portfolio monitoring. Formal RICS valuation required for lending, legal, or regulatory purposes — Propra can instruct RICS valuer via panel API

K.2 Integration requirements

- ATTOM AVM API — US commercial property automated valuation
- CoStar AVM / Analytics API — US and UK commercial AVM
- CoreLogic Property Data API — US valuation and risk data
- CBRE Valuation API (where available via data partnership) — institutional grade AVM reference
- RICS registered valuer panel — for formal instructions (API instruction, same as surveyor panel)

L Complete Updated Screen List

All screens including additions from this addendum

Group	Screen	Route	New / updated in this addendum
PORTFOLIO	**Value Dashboard**	`/dashboard`	Zero-touch: all KPIs auto-populated from APIs on property add
PORTFOLIO	**All Properties**	`/properties`	Satellite thumbnail auto-loaded for each asset
PORTFOLIO	**Portfolio Analytics**	`/analytics`	AVM-based valuation history chart added
PORTFOLIO	**Mapping & Site View**	`/map`	**NEW — interactive multi-layer map for all portfolio assets**
INCOME	**Rent Optimisation**	`/rent`	Leverage score + AI renewal strategy + draft letter — fully automated
INCOME	**Lease Restructuring**	`/leases`	AI heads of terms generation, digital signing workflow
INCOME	**Ancillary Revenue**	`/ancillary`	5G/EV/Solar application submission added — not just ID
INCOME	**AI Opportunities**	`/opportunities`	Fully ranked by live calculated saving, not estimated
COST	**Insurance Audit**	`/insurance`	Live binding quotes from carrier APIs — no illustrative data
COST	**Utility Switching**	`/energy`	Live tariff switch execution via supplier APIs
COST	**CAM Recovery**	`/cam`	Reconciliation statement auto-generated, recovery letter via AI
COST	**Tax & Compliance**	`/tax`	Rates liability auto-calculated, SDLT from HMRC tables
ASSET	**Hold vs Sell**	`/hold-sell`	AVM-based valuation, live cap rate comparables
ASSET	**Acquisitions Scout**	`/acquisitions`	AI underwriting + income opportunity on every deal card
ASSET	**Refinance Centre**	`/refinance`	Live SOFR, automated DIP to lender panel
ASSET	**Valuation**	`/valuation`	**NEW — AVM per property, monthly refresh, valuation history**
PLANNING	**Planning & Dev**	`/planning`	EXPANDED — PD rights, AI appraisal, renderings, application submission
LEGAL	**Transaction Room**	`/transactions`	**NEW — full deal room for sale/purchase/refi/lease**
LEGAL	**Legal & Searches**	`/legal`	**NEW — automated search ordering, AI lease drafting, panel instruction**

OPS	Rent Clock	/rent-clock	AI renewal letter + digital sending workflow
OPS	Tenants	/tenants	Tenant portal access, payment tracking, correspondence log
OPS	Work Orders	/work-orders	Contractor tendering via API, structured quotes, digital instruction
OPS	Compliance	/compliance	One-tap certificate ordering from vetted surveyor panel API
OPS	Project Intelligence	/projects	NEW — CapEx planning, project P&L, BCIS benchmarking
PLATFORM	Marketing Suite	/marketing	NEW — brochure + IM generator, satellite imagery, AI narrative
PLATFORM	Opportunity Inbox	/inbox	Unified queue — all savings calculated from live data
PLATFORM	Ask Propra AI	/ask	Full portfolio context, per-screen prompts, action execution
PLATFORM	Settings	/settings	Integration management, API key status, team access

M Complete API & Service Integration — Addendum

Additional integrations not covered in v3.0

Service	Category	Market	Purpose + notes
Google Maps Platform	Mapping + satellite	Global	Satellite imagery, Street View, geocoding, distance matrix, places. Static Maps API for brochures. Maps JS for interactive. Standard pay-per-use pricing.
Mapbox GL JS	Mapping	Global	Customisable interactive maps, planning layer overlays, boundary polygons. Alternative/supplement to Google Maps.
Google Maps Solar API	Solar feasibility	Global	Roof-level solar potential: annual sunshine, usable area, estimated generation kWh. Free tier available.
OS National Geographic Database	Boundaries + mapping	UK	Property boundaries, address points, building footprints, use class. OS Data Hub API.
Environment Agency Flood API	Flood risk	UK	Flood zones 1/2/3, historical flooding, flood alerts. Public API, free.
FEMA NFHL API	Flood risk	US	National Flood Hazard Layer, flood zone by address. Public API, free.

Stable Diffusion API / DALL-E API	Rendering visuals	Global	AI-generated planning visualisations conditioned on Street View and proposed scheme description. Stability AI or OpenAI.
Planning Portal API	Planning	UK	Application submission, status, decisions. Official UK Planning Portal.
Search Acumen API	Conveyancing searches	UK	Local authority, environmental, water, drainage searches — ordered and returned digitally.
TM Group API	Conveyancing searches	UK	Alternative search provider with API — same search types.
InfoTrack API	Conveyancing + legal	UK	Property searches, Land Registry AP1/TR1 submission, SDLT returns.
DocuSign API	Digital signatures	Global	NDA signing, lease execution, legal documents. REST API, well-documented.
HelloSign / Dropbox Sign	Digital signatures	Global	Alternative to DocuSign — similar capability, different pricing.
GoCardless API	Rent payments	UK/EU	Direct debit collection for commercial rent — variable direct debit for rent reviews.
Stripe API	Payments	Global	Card payments, ACH (US), international. Fallback to GoCardless for UK DD.
TrueLayer API	Open Banking	UK/EU	Bank account connection, payment confirmation, rent receipt verification.
Plaid API	Open Banking	US	US bank account data, payment confirmation, accounting reconciliation.
Adobe PDF Services API	PDF generation	Global	Brochure and document generation, PDF assembly, stamping, splitting.
BCIS API	Construction costs	UK	Building Cost Information Service — cost benchmarks by trade, specification, region.
Groundsure Avail API	Environmental searches	UK	Contamination, ground stability, flooding, energy — single API.
CTIL / Cornerstone API	5G mast data	UK	Site suitability for telecoms infrastructure, market rent benchmarks.
Companies House API	Company data	UK	Director data, accounts, credit events, insurance verification. Free public API.
D&B Hoovers API	Tenant credit	Global	Business credit reports, payment behaviour, financial health scores.
Creditsafe API	Tenant credit	UK/EU	Business credit scoring, CCJ data, payment history.
Ofcom Sitefinder API	Mobile coverage	UK	Mobile network coverage by location and operator — for 5G mast feasibility.
Glenigan API	Construction intel	UK	Planning pipeline, project data, contractor activity by sector and geography.

PROPRA — FULL AUTOMATION SPECIFICATION

No phone calls. No emails to brokers. No waiting 5 days. No manual data entry.

Every quote is machine-generated from a live API. Every switch is executed by software. Every document is AI-generated. Every notification is automated. Every saving is calculated from real data returned in real time.

If a feature requires a human at Propra to do anything, it is not finished.